Database Management System (DBMS)

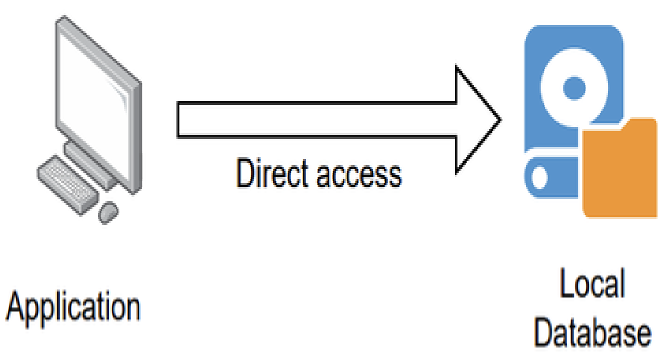
Schema :- Schema means Logical Representation (मतलब कोई भी entity का table बनाने से पहले मन में सोचना i.e. logically)

DataBase Architecture = 3 types

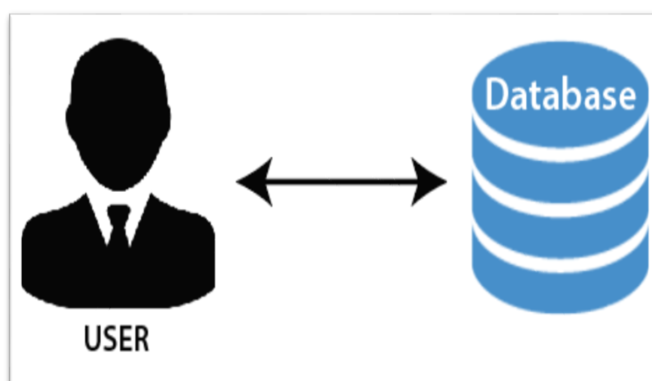
1. Single Tier architecture
2. Two Tier architecture
3. Three Tier architecture

1.Single Tier Architecture

- इस architecture में data(file,image,video,....) ये सब अपने system के ही storage device में मौजूद हो, उसे 1टियर architecture कहते हैं |
- Database is directly available to the user.
- 1-Tier architecture is used for local application.
- Any request made by client doesn't require a network connection to perform the action on the database.



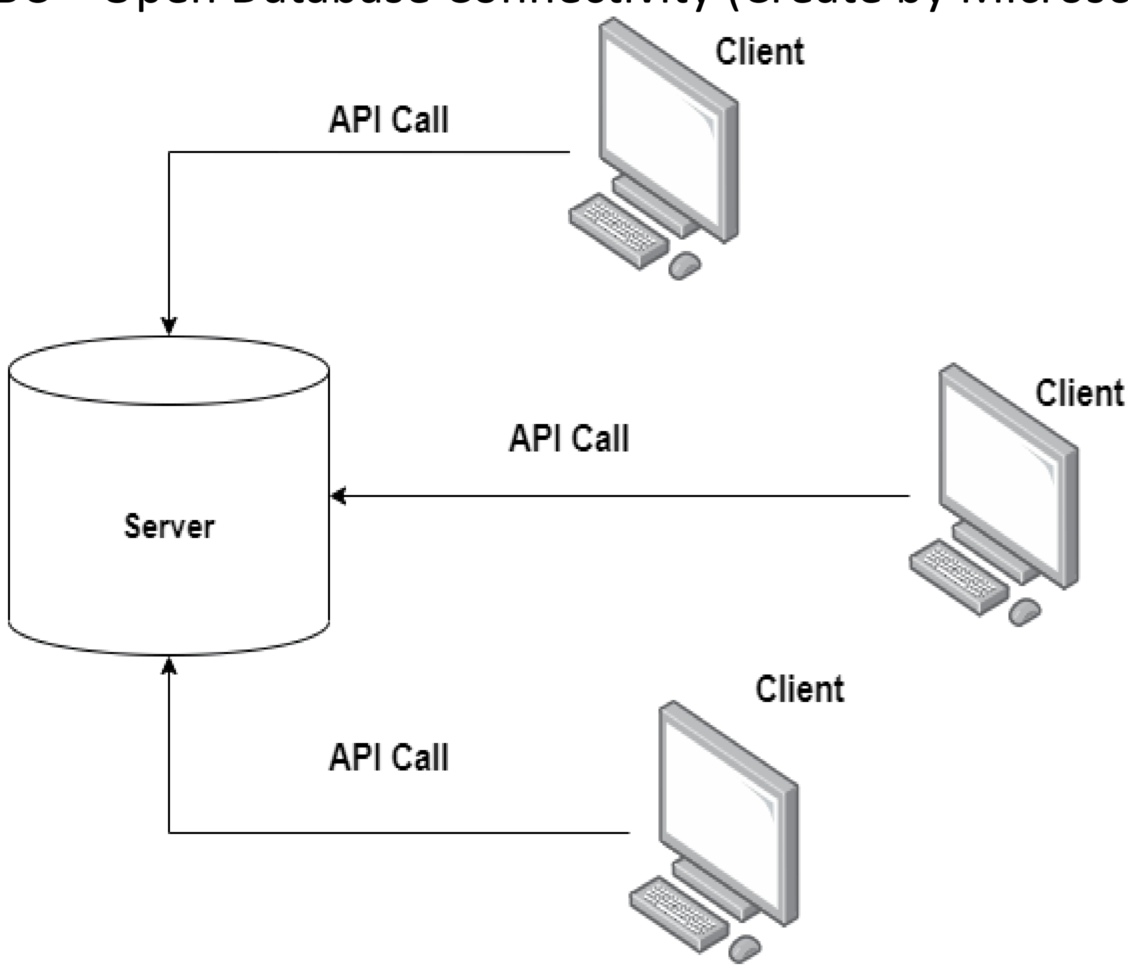
1 - Tier Architecture



1-tier Architecture

2. Two Tier architecture (2-Tier)

- Database system is present at the server machine.
- Whenever client machine makes a request to access the database present at server using a query language like SQL. (मतलब user को जब database की जरूरत होगा तब वो सर्वर से request करेगा query language के द्वारा)
- It Includes an Application Layer b/w user and DBMS.
- JDBC & ODBC are used for the interaction b/w Server & Client based on API (Application programming interface).
- JDBC = Java Database Connectivity (Create by Sum Microsystem)
- ODBC = Open Database Connectivity (Create by Microsoft)



3 Three Tier architecture

:- This architecture has three levels (3 Levels)

i. **External Level / View Level / Client Level / Upper Level / User Level / Level 1 (कुल 6 नाम)**
ii. **Conceptual Level / Logical Level / Server / Level 2**
iii. **Internal Level / Physical Level / Database / Level 3**

i. External Level

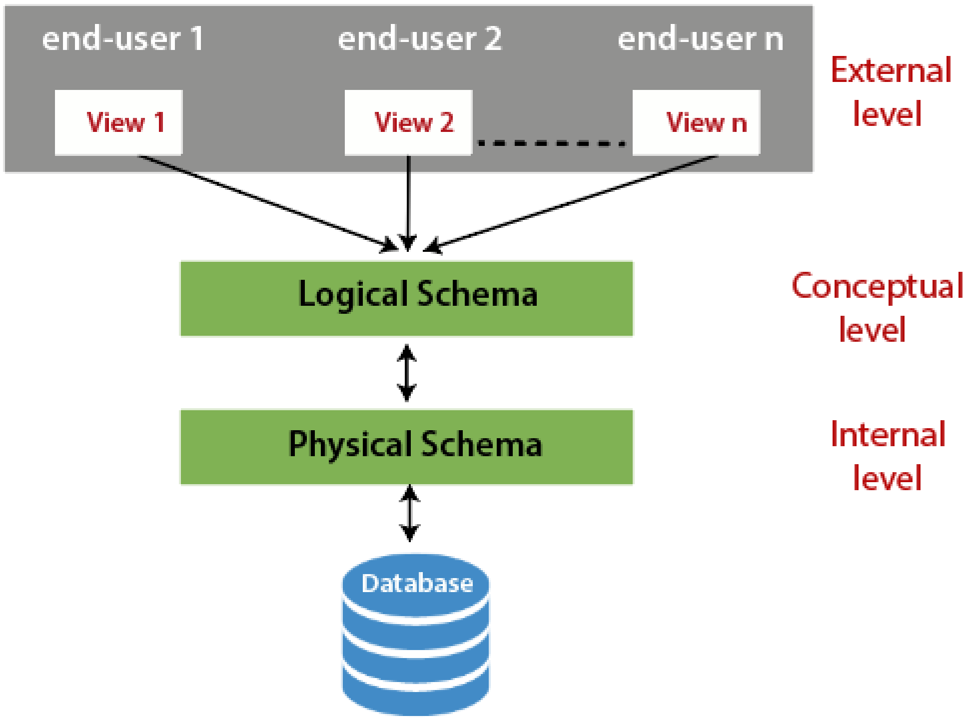
- This layer is closest to the user.
- User doesn't need to know the database Schema
- User only concerned about data which is what returned back.

ii. Conceptual Level

- Describes the Structure of the whole database.
- The Conceptula level doesn't care for how the data in the database is actually stored.
- Database constraints and security are also implemented in this level of architecture.
- Programmers and database administrators work at this level.

iii. Internal Level

- This level describes how data is actually stored in the database.
- Data is stored in the external hard drives in the form of bits or data is stored in files and folders.
- This level discusses Compression and encryption techniques.



DBMS Architecture

❖ SQL (Structured Query Language)

:- SQL is a language to operate database

➢ History of SQL

- SQL appeared – 1974
- SQL Developed by – IBM Corporation
- Father of SQL – Donald D. Chamberlin & Raymond F. Boyce
- SQL approved by ANSI – 1986
- SQL approved by ISO – 1987

ANSI (American National Standards Institute

ISO (International Organization for standardization)

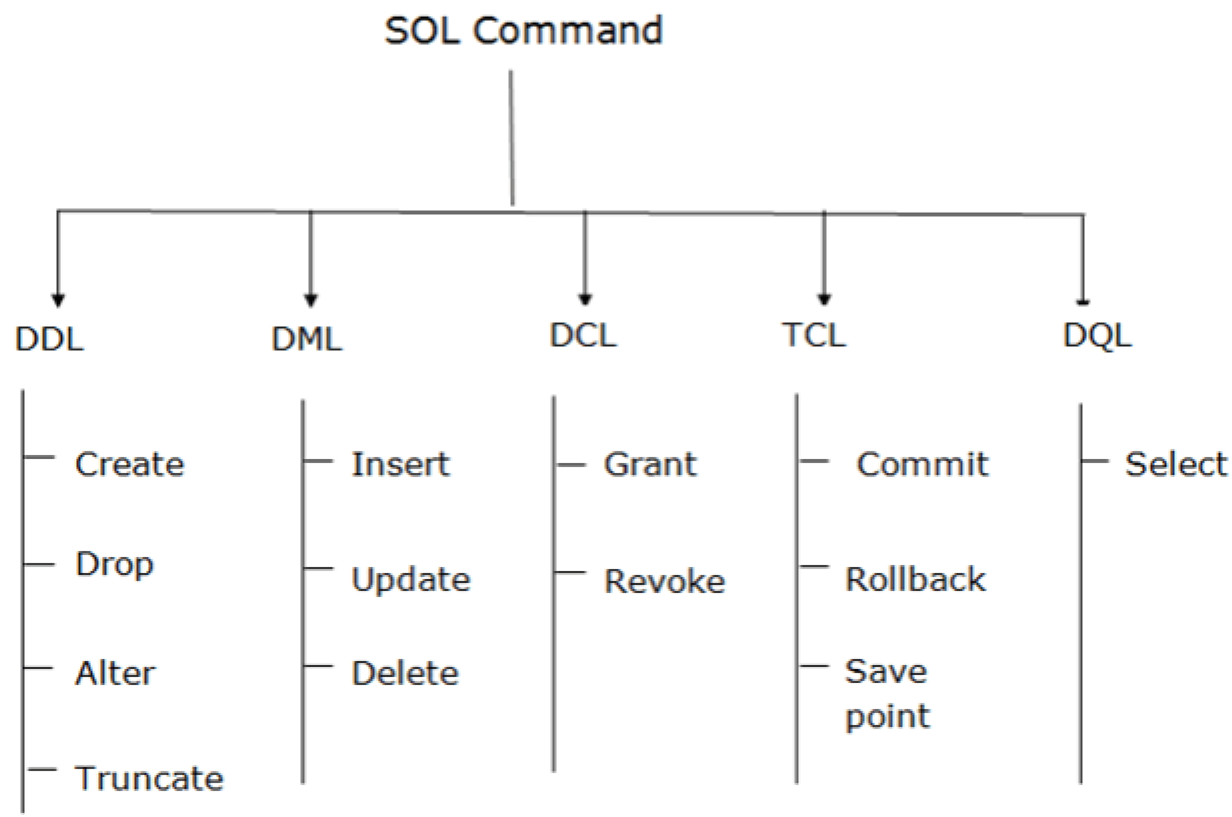
- SQL comes in Generation – 4^{th} Gen
- SQL was developed by = IBM Corporation
- Old name of SQL = SEQUEL (Structure English QUEry Language)
- SQL server is owned and developed by Microsoft Corporation.
- SQL was first Commercially distributed by – Oracle
- Structure query language is = Not Case sensitive
- Every SQL statement should ends with a = Semicolon;

➢ SQL Advantage = 6

i. High speed
ii. No coding needed
iii. Well defined standards = ISO and ANSI
iv. Portability = Used in laptop, PC, mobile phones
v. Interactive language = Interact with database
vi. Multiple data view (see/edit/rename,… give permission)

❖ SQL Commands & Language

Database में data को SQL के सहयता से Create, Modifiy and Retrive करते हैं, जिसके लिए कुछ command और language की जरूरत होती हैं |



➢ Types of SQL Language

1. DDL (Data Definition Language)
2. DML (Data Manipulation Language)
3. DCL (Data Control Language)
4. TCL (Transaction Control Language)
5. DQL (Data Query Language)

Red Color is = Structure & Other is record

No.	Name	Age	Department	DOA	Fee	Gender
1	Piyush	24	Computer	10/01/97	120	Male
2	Shivam	21	History	24/03/98	200	Male
3	Sudha	22	Hindi	12/12/96	300	Female
4	Surya	25	History	01/07/99	400	Male
5	Rohit	22	Hindi	05/09/97	250	Male
6	Mohit	30	History	27/06/98	300	Male
7	Sneha	34	Computer	25/02/97	210	Female
8	Yash	23	Hindi	31/07/97	200	Male

Record

value

column

Practical use open :- google – Type – Oracle Live →
https://www.oracle.com/database/technologies/oracle-live-sql/

1. DDL Command

i. **Create :-** Create a New Table

ii. **Drop :-** Permanently Delete Structure & Record

iii. **Truncate :-**

iv. **Alter :-** Change Structure (क्या - क्या change होगा)

a) Add Columns (Above Fig. Add new column Marks)

b) Remove Columns (Remove Age column)

c) Modifiy Data type (Now – No में 1,2,3,.... है जो की int है, यदि हम no ज्यादा हो गया या जरूरत पड़ तो C001,C002,.... varchar data type use कर सकते हैं)

d) Modifiy Data type length

e) Add & Remove Constraints

f) Remove Columns/table

v. **Rename :-** Rename Structure (गलत को सुधारना,change नहीं करना)

2. DML (Data manipulation language)

I. **Delete :-** Remove/Delete one or more Row use where condition.

Table = student

No.	Name	Age
1	Piyush	24
2	Shivam	21
3	Sudha	22
4	Surya	25

Delete = DML

Syntax :- Delete from Student where no=1,2;

For check - Desc Student; output →

No.	Name	Age
3	Sudha	22
4	Surya	25

Drop = DDL

Syntax :- Drop Student; output → drop Success

For check - Desc Student; output → no student table found

Turncate = DDL

Syntax :- Turncate Student;

For check - Desc Student; output →

No.	Name	Age

II. **Update :-** update/modify value of a column.(not change structure)

III. **Insert :-** insert data

3.DCL (Data control Language)

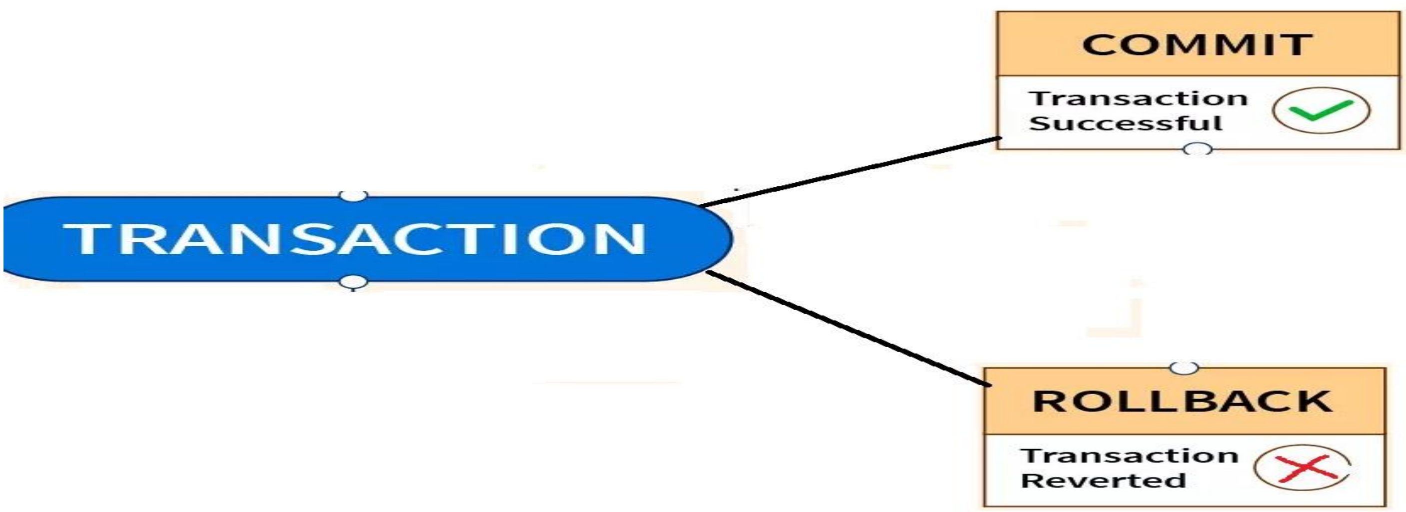
- मतलब data का कण्ट्रोल किसके पास हैं |

i) Grant :- Data का control / Permission user को देना i.e. Grant

ii) Revoke :- Data का control / Permission user से वापस लेना i.e. Revoke

4. TCL (Transaction Control Language)

लेने -देना कैसे हो रहा हैं |



i) **Commit :-** Permanent Save all transaction is Successful

ii) **Rollback:-** Undo if transaction not save in database i.e. failed.

iii) **Save Point :-** Creates points within the groups of transactions in which to ROLLBACK. i.e. Transaction process.

5.DQL (Data Query Language)

i) Select :- The Select command is used to select data form a database.

➢ Some important SQL command = 11

1. **SELECT** → Extracts data from a database
2. **UPDATE** → Updates data in a database
3. **DELETE** → Deletes data form a database
4. **INSERT INTO** → Inserts new data into a database
5. **CREATE DATABASE** → Creates a new database
6. **ALTER DATABASE** → modifies a database
7. **CREATE TABLE** → Creates a new table
8. **ALTER TABLE** → Modifies a table
9. **DROP TABLE** → Deletes a table
10. **CREATE INDEX** → Creates an index (Search key)
11. **DROP INDEX** → Deletes an index

➢ SQL Query Order

1. **SELECT** → fields / Columns
2. **FROM** → tables / views
3. **WHERE** → conditions / Expression
4. **GROUP BY** → fields
5. **HAVING** → Conditions on aggregate fields
6. **ORDER By** → fields

✓ DISTINCT = Use Remove the duplicate rows from the result.

✓ SQL keywords are not case sensitive

✓ In SQL query order only the SELECT and FROM clauses are required. All the other clauses are optional.

➢ SQL Operators

i. Airthmetic operators

ii. Bitwise Operators

iii. Comparison Operators

Student

No.	Name	Age	Department	DOB	Fee	Gender
1	Piyush	24	Computer	10/01/97	120	Male
2	Shivam	21	History	24/03/98	200	Male
3	Sudha	22	Hindi	12/12/96	300	Female
4	Surya	25	History	01/07/99	400	Male
5	Rohit	22	Hindi	05/09/97	250	Male
6	Mohit	30	History	27/06/98	300	Male
7	Sneha	34	Computer	25/02/97	210	Female
8	Yash	23	Hindi	31/07/97	200	Male

✓ Total fields/structure in this fig is = 7 (no, name, Age, Dept.,DOA,Fee,Gender)

Q1. In above fig. Fetch करे the No,Name and Department fields को of the student table.

Syntax :- SELECT Fields name,…..From table_name;
Example :- SELECT No,Name,Department From Student;

Q2. Fetch all fields of the Student table
Syntax :- SELECT * From table_name;
Example :- SELECT* From Student;

Q3. Fetch the No, Name and Fee fields from the Students table, where the fee is greater than 300.
Syntax :- SELECT Field_name,…….From Table_name WHERE Field_name > condition;
Exp :- SELECT No,Name,Fee FROM STUDENT WHERE FEE > 300;

Q4. Fetch the No,Name and Fee fields from the students table for a student with the name mohit.
Syntax :- SELECT No,Name,Fee From Students Where Name ='mohit';

Q5. Fetch the No,Name and fee fields from the Students table, where the fee is greater than 210 and the age is less than 30 years.
Example :- SELECT No,Name,Fee From Student Where Fee >210 And age <30;

Q6. Write a query, which will DELETE a customer, whose No is 8.
Example :- DELETE FROM Student Where No = 8;

Q7. Write a query, orders all rows from students in ascending order by age.

Example :- SELECT* From Students Order by age ASC; OR
SELECT* From Students Order by age;

If Descending order by age

Example :- SELECT* From Students Order by age DESC;
SELECT* From Students Order by age (XXXXXXX Wrong)

Q8. Write a query, → Select name and age of students who don't take a History → and sort them by name in descending order.

Example :- SELECT name,age FROM students WHERE NOT Department = 'History' Order by name DESC;

❖ POSTGRE SQL

- Postgres Structure query language.
- PostgreSQL also known as = Postgres.
- Postgre SQL is = Free & Open source RDBMS
- Postgre SQL is used as Data warehouse for web applications, mobile application
- Postgre SQL is an Advance Version of SQL
- It is written in = C language
- It is used by = uber, Netflix, Instagram, Spotify etc
- Postgre SQL is open source
- Postgre SQL is Object RDBMS
- Support for SQL was added in = 1994

- Released as Postgres95 in 1995.
- Postgre SQL offers data types to store = IPv4, IPv6 and MAC Address.
- In Postgre SQL a named collection of tables is called = Schema
- Postgre SQL used Client/Server model of Communication.
- Support for JSON = JavaScript Object Notation.
- Maximum length of a table name is PostgreSQL = 64 Characters.

❖ NOSQL Database

- SQL is used when we have structured data in the form of tables, if all data are stored in Unstructured manner in that case we use NoSQL database. (SQL का use हम structure data के लिए करते हैं ,यदि हमारे पास data Unstructure form में हो तो NoSQL का प्रयोग करते हैं |)
- Benefit of NoSQL = Easy Schema Evolution
- MongoDB = Handle Complex queries / handling large volumes of data
- NoSQL is hierarchical in nature.
- No any support to ACID properties
- Distributed Database.

No	RDBMS s/w OR SQL s/w	NoSQL s/w
1	PostgreSQL	MongoDB
2	MySQL	DynamoDB
3	Microsoft SQL server	Cassandra
4	SQLite	CouchDB
5	SQL Server Express	CouchBase
6	Sequel Pro	Neo4J
7	MariaDB	Amazon SimpleDB
8	Db2 Express-C	HBase
9	CUBRID	
10	Firebird	